

Estructura de carpetas propuesta

AllRight/

```
├── .github/
|   └── workflows/
|       ├── ci.yml          # Lint, test, build, docker
|       └── cd.yml          # Deploy agnóstico (compose / registry)
|
├── apps/
|   └── api/                # NestJS bootstrap (capa Presentation)
|       ├── src/
|       |   ├── main.ts
|       |   ├── app.module.ts
|       |   ├── controllers/    # HTTP adapters
|       |   ├── guards/
|       |   ├── filters/
|       |   ├── interceptors/
|       |   └── metrics/        # prom-client registry + /metrics
|       ├── Dockerfile
|       ├── nest-cli.json
|       └── tsconfig.json
|
├── packages/
|   ├── domain/            # Capa Domain (sin dependencias externas)
|   |   └── src/
|   |       ├── entities/      # User, AuditLog, GeofenceZone, LocationProof
|   |       └── value-objects/  # Coordinates, ProofHash, Email
```

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| | | └─ ports/          # Interfaces (repositorios, ZKP, cache)
| | | | └─ user.repository.port.ts
| | | | └─ audit.repository.port.ts
| | | | └─ location-proof.port.ts
| | | | └─ cache.port.ts
| | | └─ services/       # Lógica de dominio pura
| | |   └─ geofence.domain-service.ts
| | |   └─ location-privacy.domain-service.ts
| | └─ package.json
| |
| └─ application/        # Capa Application (use cases)
| | └─ src/
| | | └─ use-cases/
| | | | └─ auth/
| | | | | └─ login.use-case.ts
| | | | | └─ register-user.use-case.ts
| | | | └─ location/
| | | | | └─ update-location.use-case.ts
| | | | | └─ verify-location-proof.use-case.ts
| | | | | └─ generate-location-proof.use-case.ts
| | | | └─ audit/
| | | | └─ log-action.use-case.ts
| | | └─ dto/
| | └─ tokens.ts        # DI tokens para ports
| | └─ package.json
| |
| └─ infrastructure/    # Capa Infrastructure (adapters)
| | └─ src/
| | | └─ persistence/

```

- | | | └─ typeorm/ # Entities ORM + mappers
- | | | └─ repositories/ # Implementan ports del domain
- | | └─ cache/
- | | | └─ redis/ # Sessions, rate-limit, GEO, concurrent users
- | | └─ zkp/
- | | | └─ circuits/ # circom (.circom)
- | | | └─ wasm/ # Compilados (gitignored, CI build)
- | | | └─ snarkjs.adapter.ts
- | | └─ observability/
- | | | └─ prometheus.module.ts
- | | | └─ metrics.service.ts
- | | └─ config/
- | | └─ env.validation.ts
- | └─ package.json
- |
- └─ web/ # Frontend (evolución de /public)
- | └─ public/
- | └─ src/
- |
- └─ e2e/ # Playwright (recomendado)
- | └─ playwright.config.ts
- | └─ fixtures/
- | └─ tests/
- | | └─ auth.mobile.spec.ts
- | | └─ location.mobile.spec.ts
- |
- └─ infra/
- | └─ docker/
- | | └─ docker-compose.yml # Local dev

```
| | └─ docker-compose.prod.yml    # Producción
| | └─ .env.example
| └─ prometheus/
| | └─ prometheus.yml
| └─ grafana/
|   └─ provisioning/
|     └─ datasources/prometheus.yml
|     └─ dashboards/dashboard.yml
|       └─ dashboards/
|         └─ allright-overview.json
|
└─ package.json          # Monorepo root (npm workspaces / pnpm)
└─ turbo.json            # Opcional: build pipeline
└─ README.md
```